

SOLIDS

- definite shape (rigid)
- definite volume

LIQUIDS

- no definite shape; able to flow
- definite volume

GASES

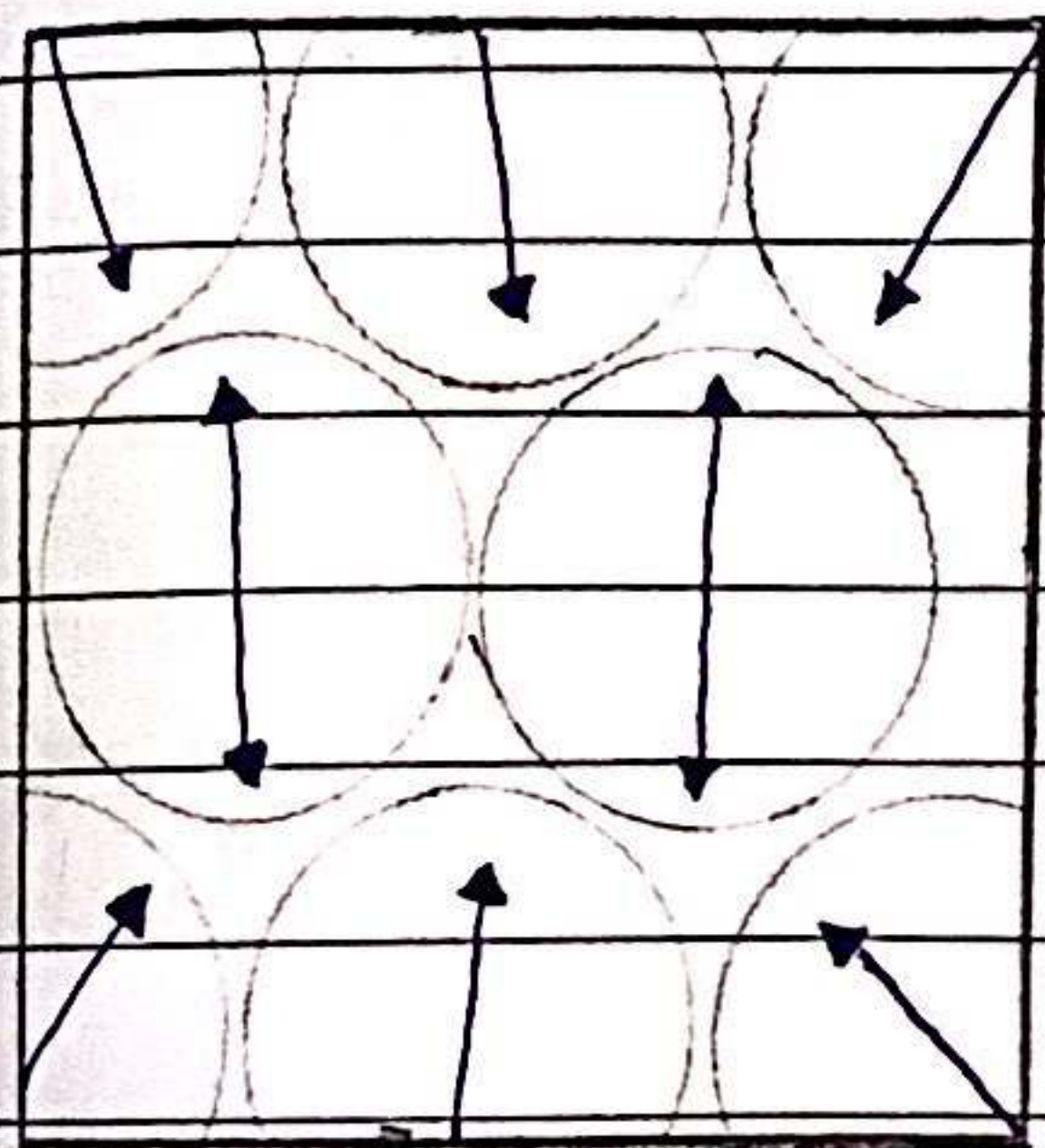
- no definite shape; will take the shape of their container
- no fixed volume; will expand to fill a closed container
- highly compressible

solid $\xrightarrow{\text{melting}}$ liquid

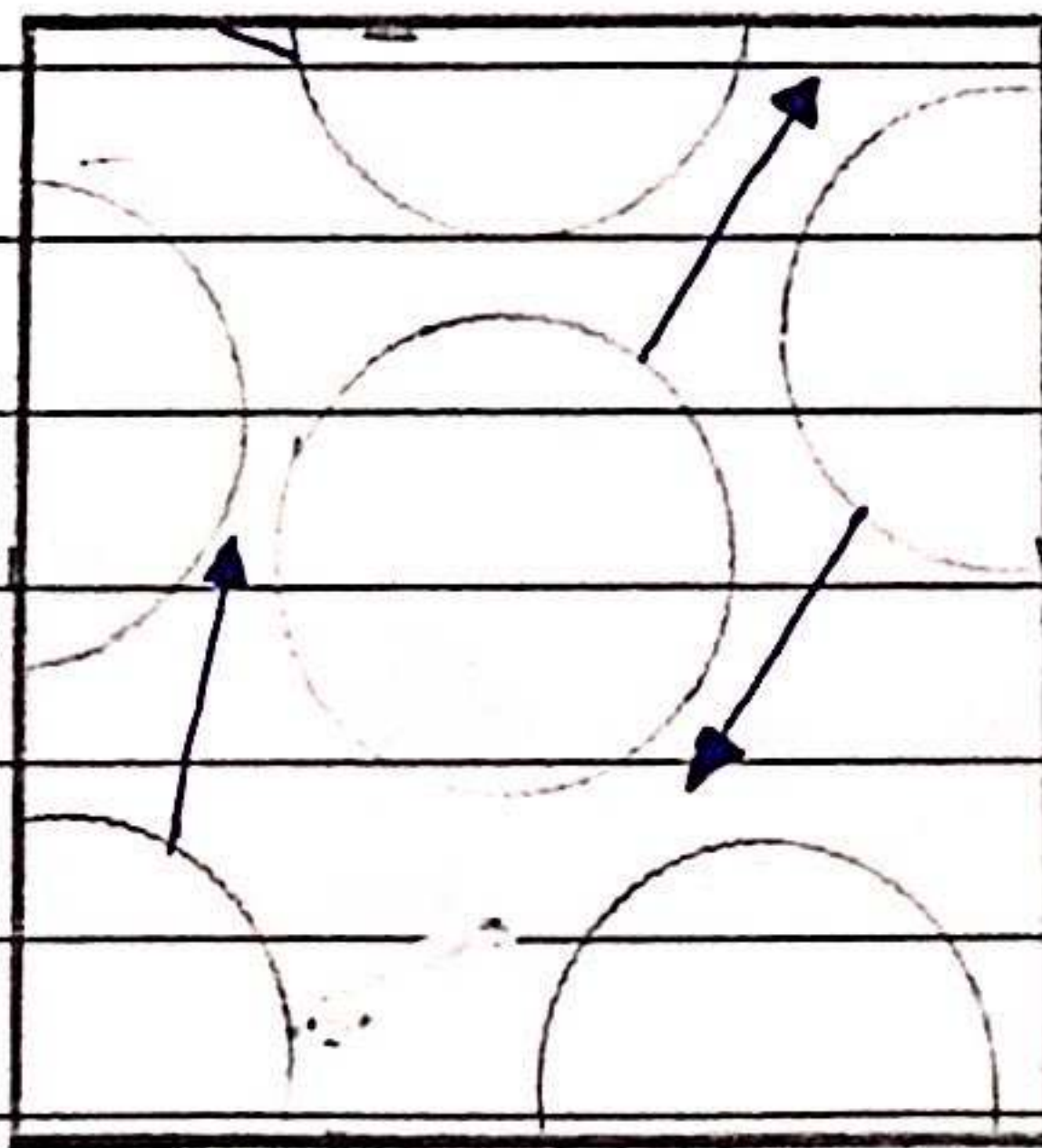
liquid $\xrightarrow{\text{freezing}}$ solid

liquid $\xrightarrow[\text{evaporation}]{\text{boiling}}$ gas

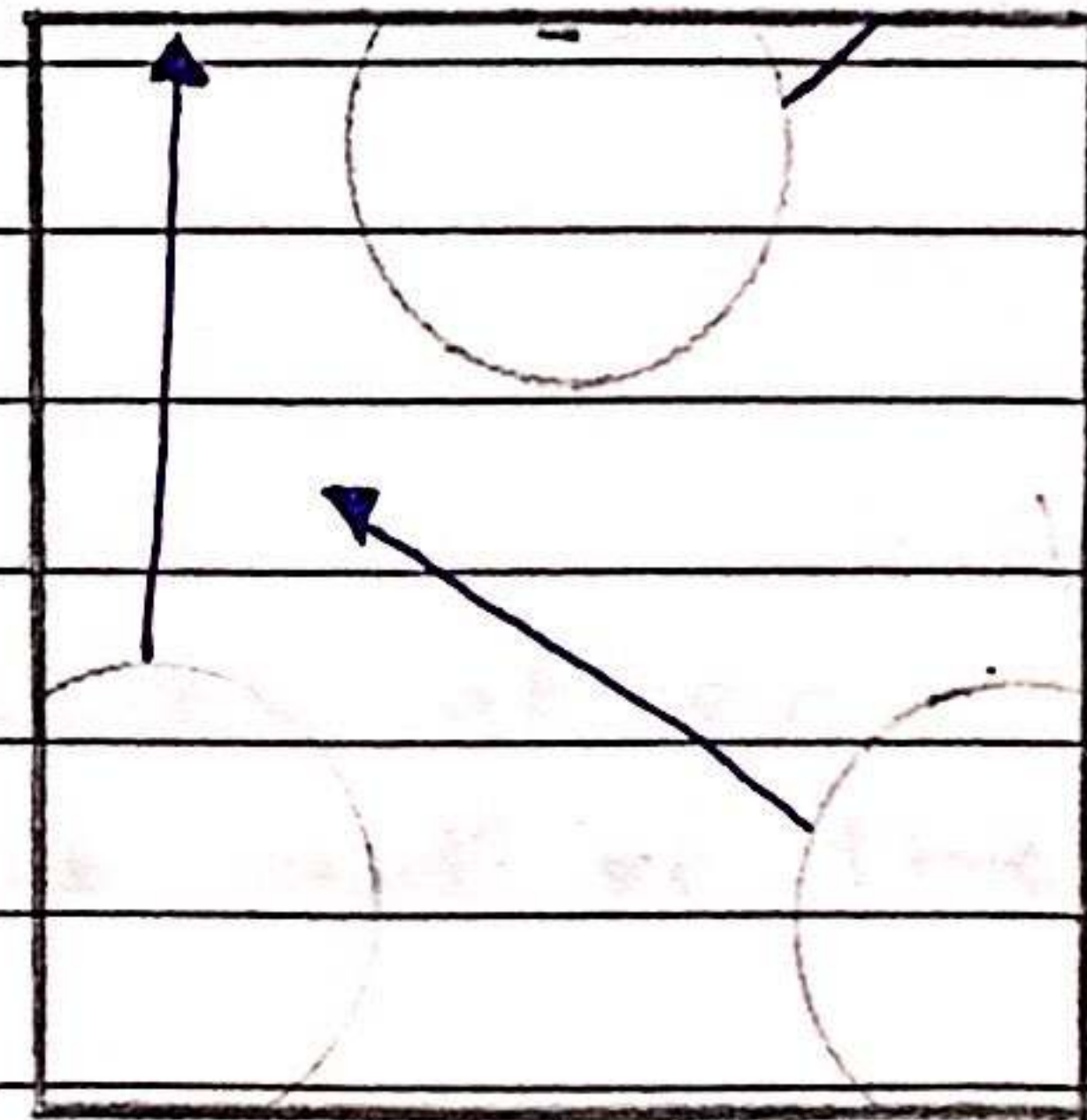
gas $\xrightarrow{\text{condensing}}$ liquid



solids



liquids



gases

in a solid...

- molecules are very close together and arranged in a regular pattern
- molecules vibrate about fixed positions

in a liquid...

- molecules are still close together but are no longer in a fixed pattern
- molecules are able to slide past each other

in a gas...

- molecules are widely separated
- molecules move about randomly at high speeds

- the speed of the molecules in a gas is proportional to the temperature and inversely proportional to the molar mass of the gas.
- the lowest kinetic energy that a particle can have is at -273°C (aka 'absolute zero')
- random motion of microscopic particles in a suspension is evidence for the kinetic particle model of matter

BROWNIAN MOTION

- the random zig-zag motion of particles after colliding with other particles or the wall of the container.
- $F = ma$ (force = mass $\times$ acceleration)

variables that affect the pressure of a gas:

- change of temperature at constant volume
- change of volume at constant temperature

- temperature $\uparrow$ [then] pressure $\uparrow$ (at a constant volume)

- volume $\downarrow$ [then] pressure $\uparrow$ (at a constant temperature)

• $pV = \text{constant}$ (pressure $\times$ volume = constant)

ex.

an unknown gas occupies a volume of 1.5 m^3
at a pressure of 100 Pa

- calculate the pressure exerted by the gas if it is compressed to a volume of 0.3 m^3
(temp. and mass of gas stay the same)

$$V = 1.5 \text{ m}^3$$

$$P = 100 \text{ Pa}$$

$$pV \rightarrow 100 \times 1.5 = 150$$

$\nearrow$ constant

$$V = 0.3 \text{ m}^3$$

$$P = ?$$

$$pV \rightarrow \frac{? \times 0.3}{0.3} = \frac{150}{0.3}$$

$$\boxed{P = 500 \text{ Pa}}$$

$$P_1 V_1 = P_2 V_2$$

$$100 \times 1.5 = 500 \times 0.3$$

Boyle's Law

converting between kelvin and degrees Celsius:

$$T (\text{in K}) = \theta (\text{in } ^\circ\text{C}) + 273$$

$$\theta (\text{in } ^\circ\text{C}) = T (\text{in K}) - 273$$

THERMAL EXPANSION

- it is a process where a body expands in either area, volume or shape when heated.

↳ Solids

- solid is heated, its particles begin to vibrate, this takes up more space and the solid expands
ex. when you can't open a tight jar
 - put it under hot water
 - the lid will expand
 - now you can open it

↳ Liquids

- liquid is heated, its particles begin to vibrate, this takes up more space and liquid expands
ex. thermometer
 - put it in something hot
 - the mercury inside will expand

↳ gases

- gas is heated, its particles begin to vibrate, this takes up more space and gas expands
ex. balloon
 - put a balloon outside on a hot day
 - the gas inside the balloon expands
 - eventually the balloon will pop

a gas expands the most, then liquids, then solids

- ↳ gases have the weakest intermolecular forces
- ↳ liquids have a stronger intermolecular force than gas
- ↳ solids have the strongest intermolecular force

everyday applications/consequences of thermal expansion

→ applications

- expansion of mercury in a thermometer
- removal of tight lids

→ consequences

- expansion of bridges / railway tracks cause them to get bent out of shape
- walls collapsing due to bulging

Internal Energy

- sum of all of the potential and kinetic energy of an object
- rise in temperature increases its internal energy
- rise in temperature increases the average kinetic energies of all the particles in the object.

Specific heat capacity

energy required per unit mass per unit temperature increase

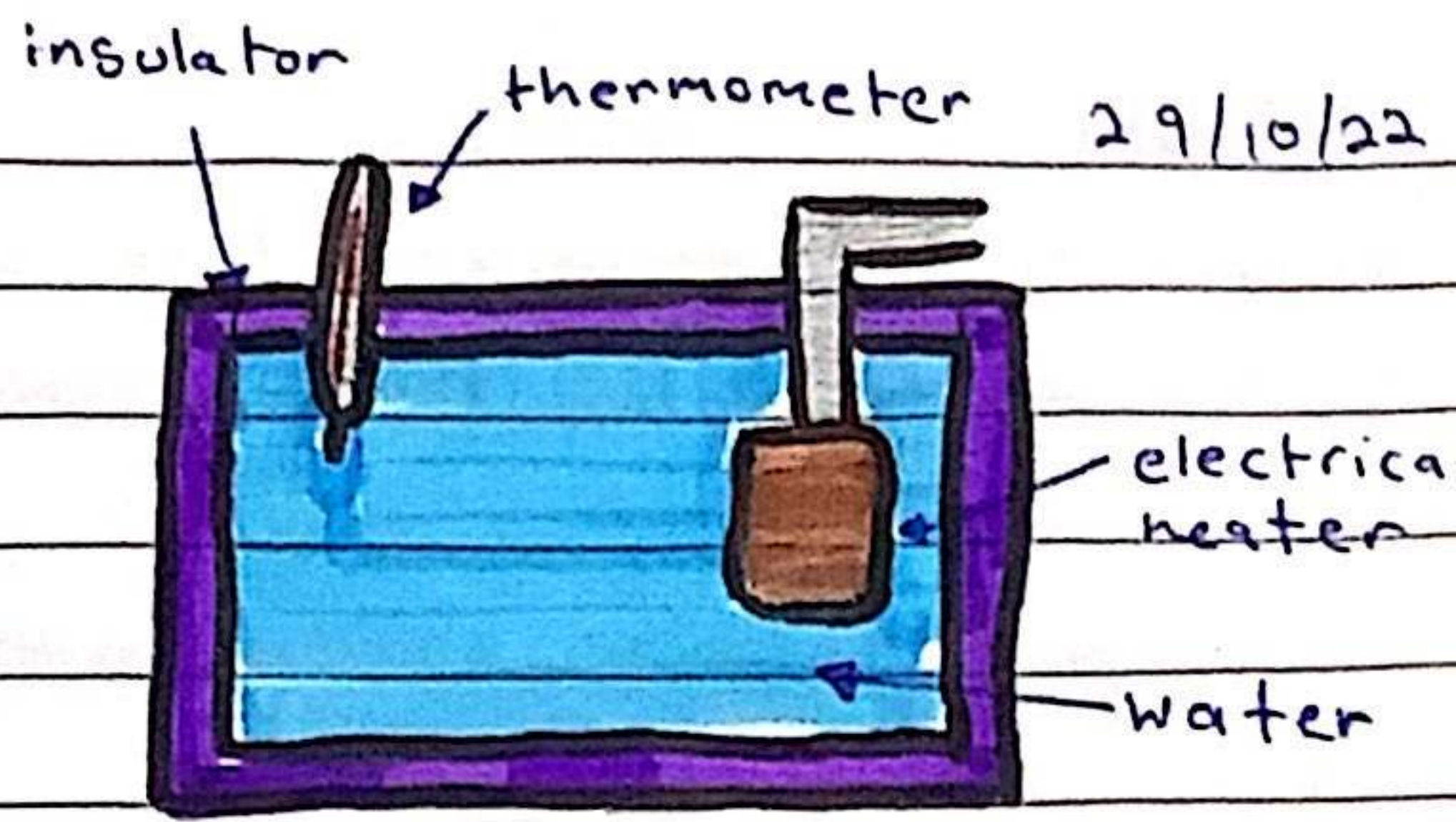
$[c = \frac{\Delta E}{m \Delta \theta}]$ e.g. 3kg of Lead → increased from 40° to 50°
→ energy transferred is 3840 J

specific heat capacity =

$$c = \frac{3840}{3 \times 10} = \frac{3840}{30} = \underline{\underline{128 \text{ J/kg}^\circ\text{C}}}$$

29/10/22

- 0.5 kg of water
- thermometer measures temperature
- elec. heater power = 50W
- heater turned on and timer turned on too
- timer is stopped when the temp. rises by 10°C



$$e = \text{power} \times \text{time} = \underline{50} \times \underset{\text{(assume)}}{420} = \underline{21000} \text{ J}$$

$$c = \frac{\Delta E}{m \Delta \theta} = \frac{21000}{0.5 \times 10} = \underline{4200} \text{ (J/Kg}^\circ\text{C)}$$

melting

- solid $\rightarrow$ liquid
- the kinetic energy rises as the temperature rises and particles begin to vibrate more rapidly
- when melting starts, there is no increase in temp. because the thermal energy supplied is used to break bonds between the solid particles, thus making it a liquid
- latent heat of fusion is the amount of energy that a solid needs in order for it to become a liquid

boiling

- liquid $\rightarrow$ gas
- the kinetic energy rises as the temperature rises and particles begin to vibrate more rapidly
- when boiling starts, there is no increase in temp. because the thermal energy supplied is used to break bonds between the liquid particles, thus making it a gas.
- latent heat of vaporization is the amount of energy that a liquid needs in order for it to become a gas.

difference between boiling and evaporation ::

- boiling occurs at a fixed temperature and throughout the liquid.
- evaporation occurs at any temperature and only on the surface.

→ melting point of water = 0°C

→ boiling point of water = 100°C

condensation and solidification ::

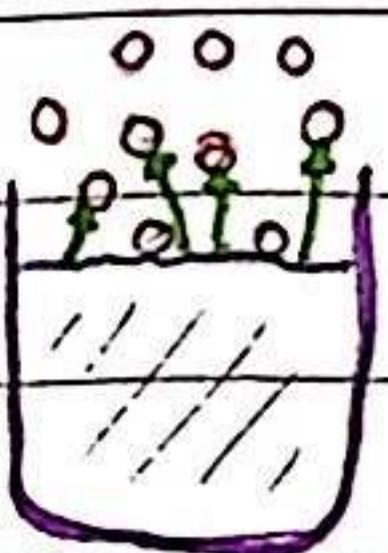
- condensation is the process by which a gas turns into a liquid ex. steam → water
- solidification is the process by which a liquid turns into a solid ex. water → ice

evaporation →

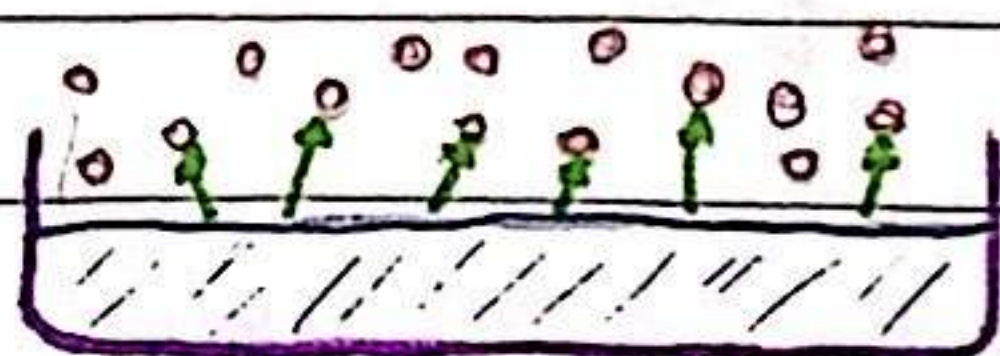
- in evaporation, the particles with more energy escape the liquid from the surface
- evaporation causes cooling of a liquid
- ↳ when the hotter particle 'breaks free' from the liquid, it takes its energy with it so the average energy of the liquid decreases, thus the liquid cools down.

factors affecting evaporation

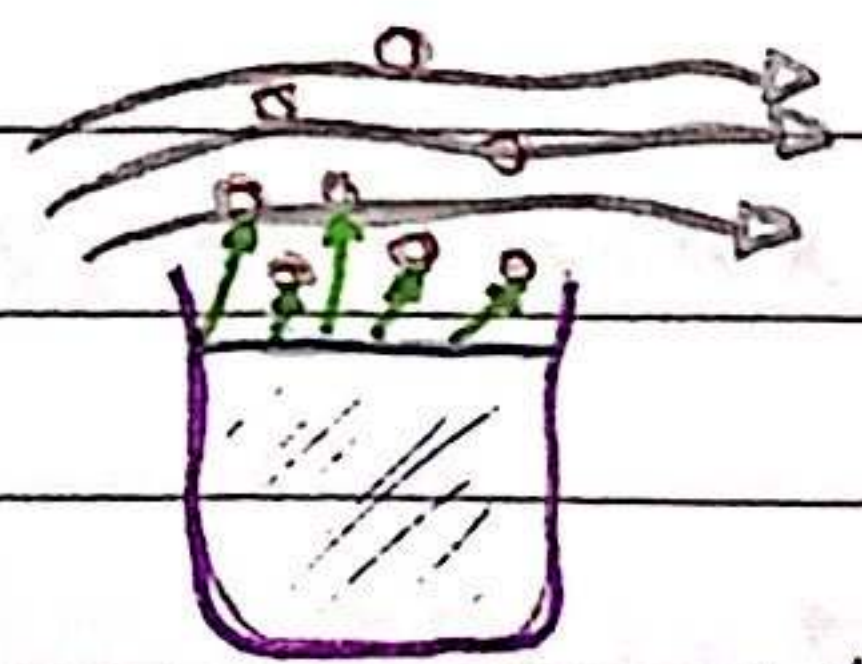
- ① temperature
- ② surface area
- ③ air movement



temperature ↑
evaporation ↑



Surface area ↑
evaporation ↑

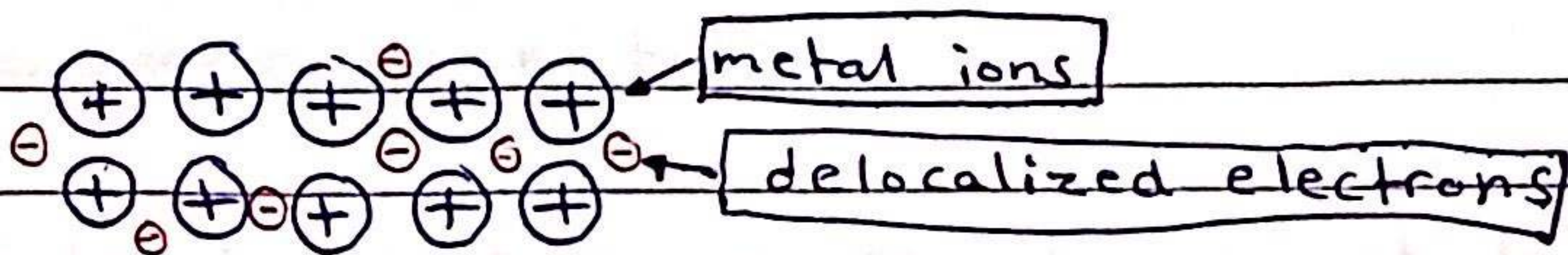


air movement ↑
evaporation ↑

conduction ::

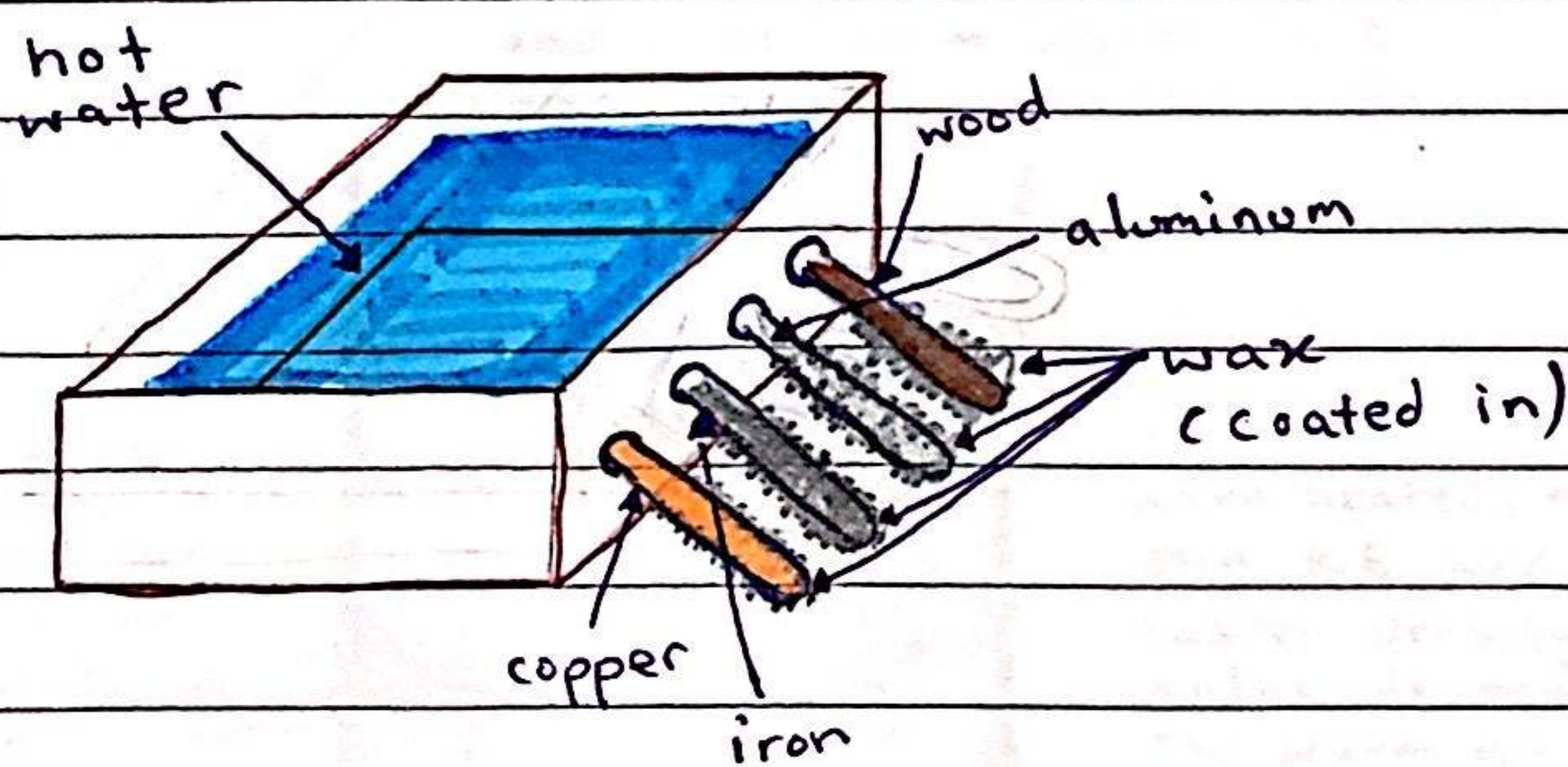
- the process by which heat or electricity is directly through the material of a substance
- occurs via the transfer of kinetic energy through the structure
- dependant on the structure of the substance

→ metals are good conductors because :-



- ↳ the electrons can quickly transmit heat/elec. by moving throughout the metal.
- ↳ when one end is heated, particles vibrate faster and neighboring particles do too.
- ↳ this creates a domino effect

experiment to demonstrate good / bad thermal conductors ::



wax melts at different extents depending on the material's conduction

- * as the heat travels through the tubes, the wax starts to melt
- * copper, iron and aluminium's wax melts the most, they are good conductors. Wood is a bad conductor because its wax barely melted at all.

- thermal conduction is faster when the particles are closer together, such as in solids, only if there are free electrons in that substance
- free electrons transfer the heat throughout
- this is why most metals are great thermal conductors
- this is also why gases and most liquids and most non-metal materials are not as good conductors as metals.

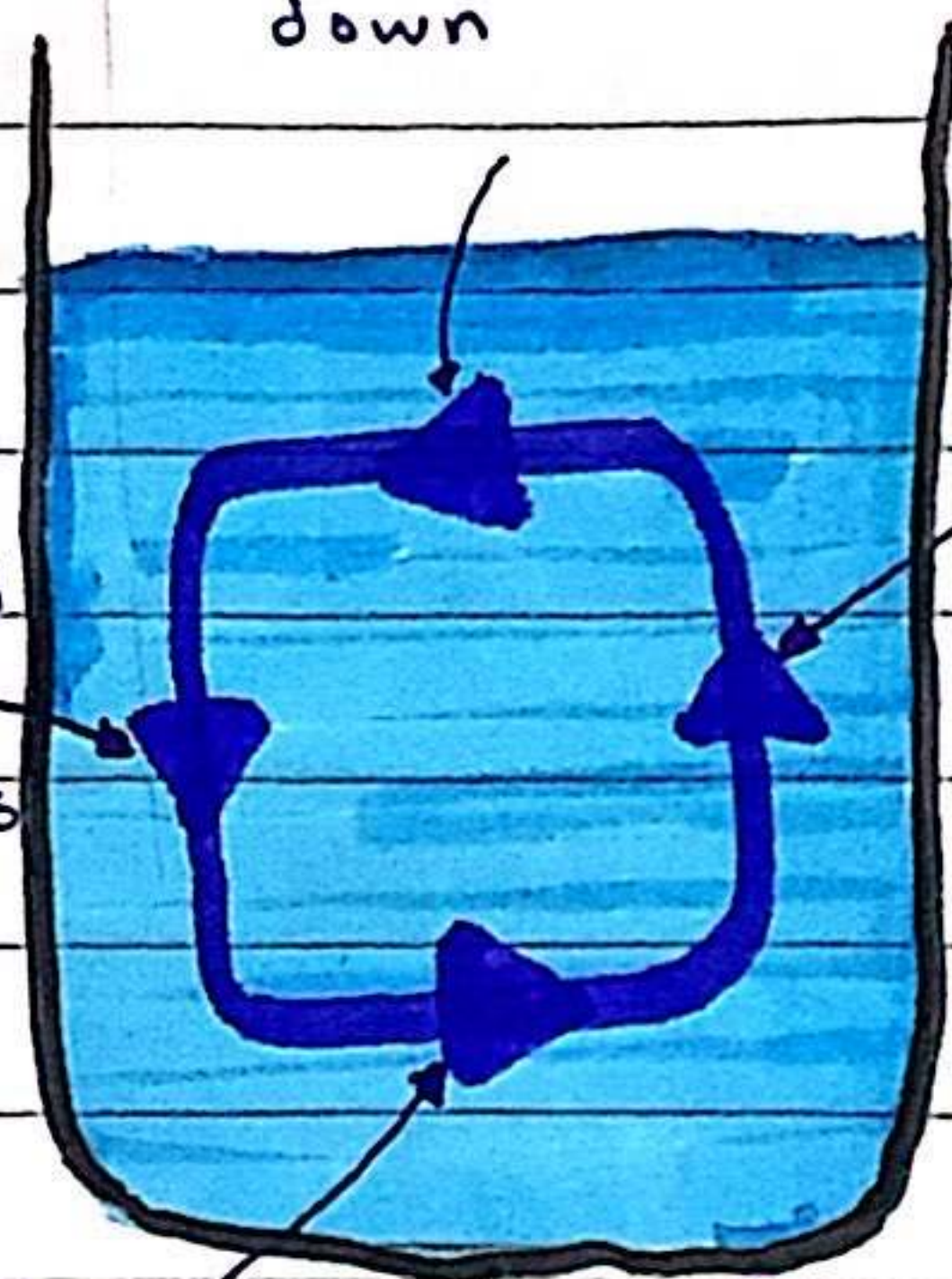
- particles in gases are too far apart to transfer energy and same with liquids
- many solids conduct thermal energy better than insulators but not as good as 'good' thermal conductors like metals.

convection:

- a method of heat transfer strictly in liquids and gases

(2) here the water cools down and more warm water rises and pushes the cold water down

(3) the water cools more and loses KE. Density of the water increases and the water sinks to the bottom



(1) when heated, molecules gain KE and move faster. Density of the water decreases. The warm water rises above the cold water

this is a cycle which repeats and creates a convection current.

(4) the water that was being heated rises as the cooler water takes its place

2/11/22

thermal radiation:

- infrared radiation that all objects emit
- thermal energy transfer by thermal radiation does not require a medium
- For an object to be at constant temperature, it needs to transfer energy
- rate of emission increases when surface temperature increases

effect of color on emission/absorption:-

- dull black is the best emitter and absorber of infrared rays
- shiny white is the worst emitter and absorber of infrared rays
- shinier the object is, the more reflective it is

* if energy in = energy out [temp. remains constant]

* if energy in > energy out [temp. increases]

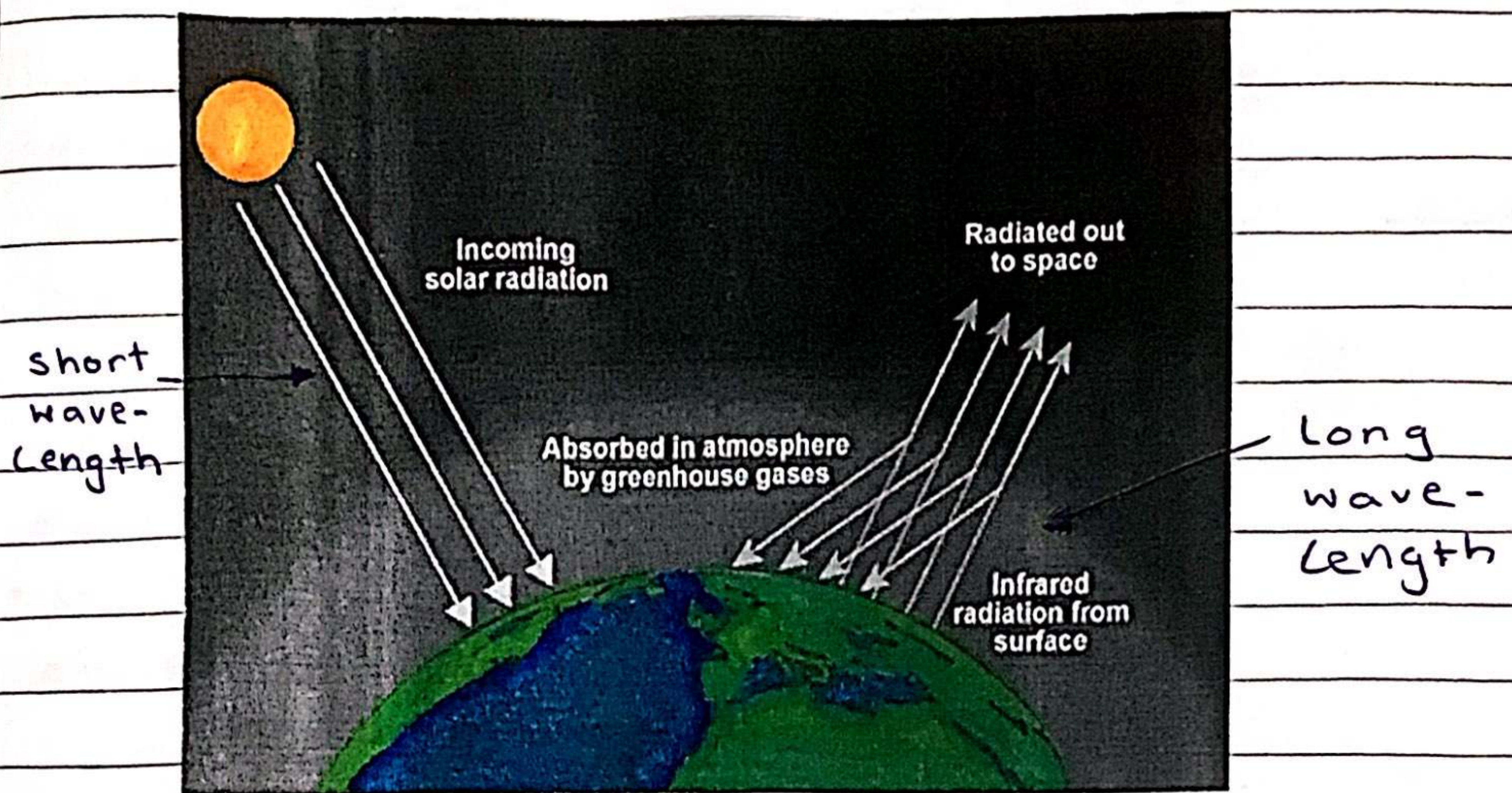
* if energy in < energy out [temp. decreases]

→ For an object to warm up, energy must be added

→ For an object to cool down, energy must be removed

Effects on radiation of colour and texture

	Matt black	White	Shiny silver
Emitting	Best	Poor	Worst
Absorbing	Best	Poor	Worst
Reflecting	Worst	Good	Best



short wave length :: (from sun)

- 25% absorbed by the ozone layer
- 75% reaches the earth's surface
↳ most radiation is converted to heat

Long wave length :: (from earth) [infrared]

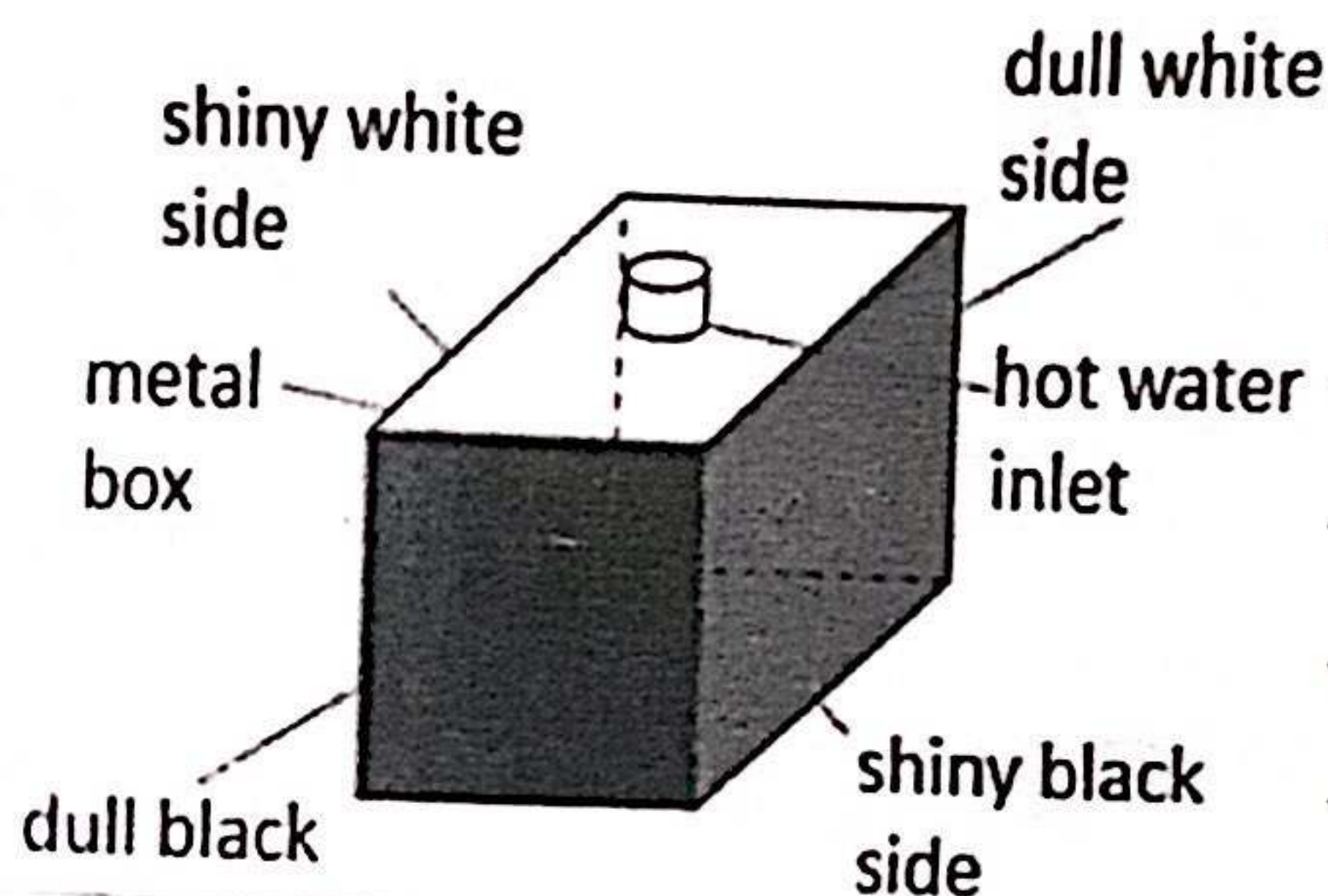
- earth's surface re-emits radiation at much longer wave lengths (infrared)
- most of the radiation is absorbed by the greenhouse gases in our atmosphere.
- that radiation is reflected back at the surface
- this is known as the 'greenhouse effect'
- because of this our planet gets hotter than usual,

examples of greenhouse gases include :-

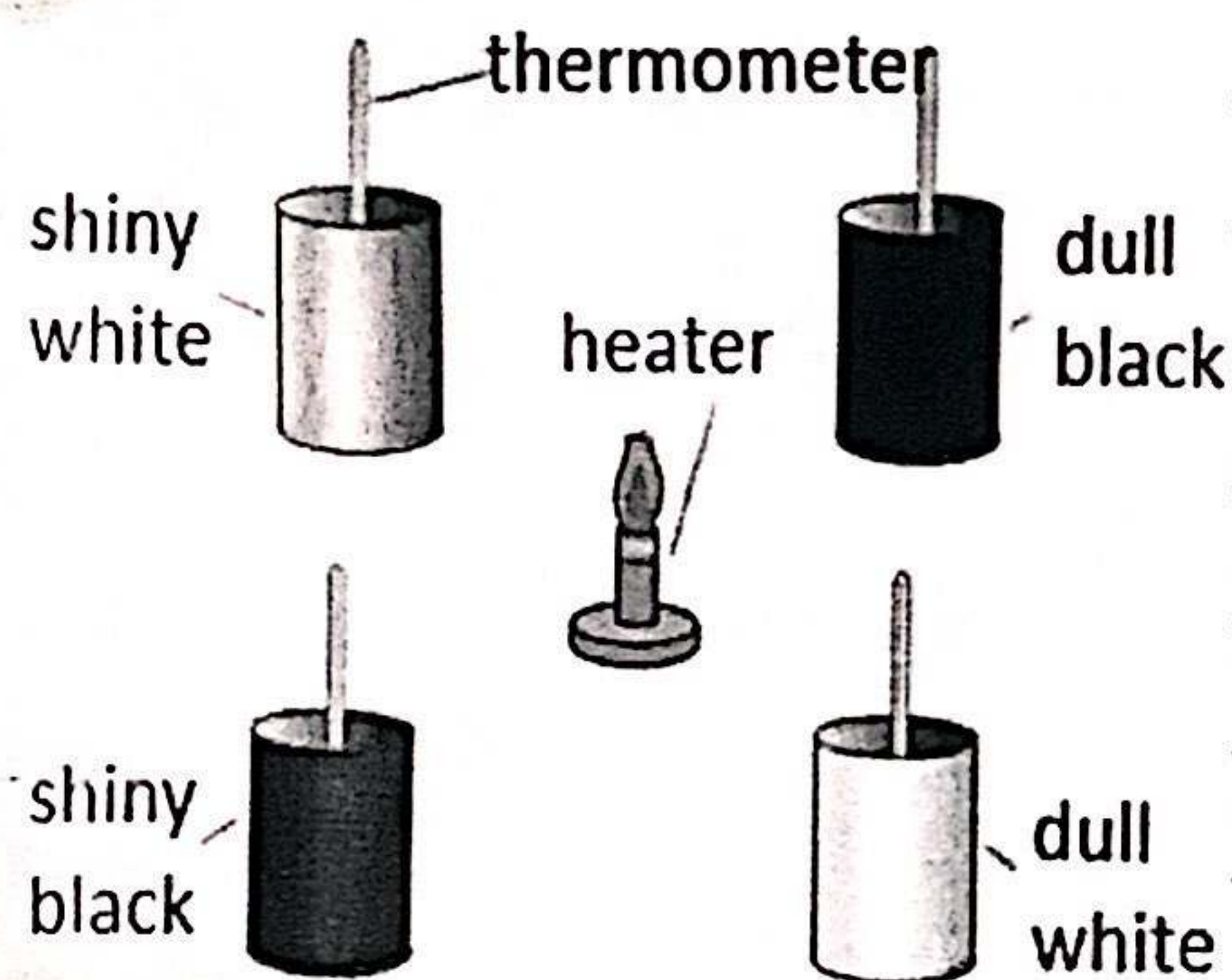
- Carbon dioxide (CO_2)
- Water vapour (H_2O)
- Methane (CH_4)
- Nitrous oxide (N_2O)
- Ozone (O_3)

distinguishing between good/bad emitters::

- place boiling water in a metal cube with different colors/textures
- measure thermal radiation emitted from each side.
- compare which surface emits the most.



distinguishing between good/bad absorbers::



- place a heater at equal distances from metal plates of different colors/textures.
- measure temperature of plates to compare which absorbs the most.

rate of emission of radiation depends on ... surface area::

- larger the surface area, more is the rate of emission
- ### surface temperature::
- more the temperature, more is the rate of emission

everyday applications of :-

conduction::

- heating a pan on a stove

convection::

- electric heaters, heating up a room

radiation::

- the light/heat from the sun

everyday consequences of :-

conduction::

- burning your hand when you touch a hot mug

convection::

- heat from the mug transferring into the air by convection currents

radiation::

- excessive solar radiation

complex uses of :-

a fire burning wood::

- heat is transferred via air currents (convection)
- nearby solid objects get heated (conduction)
- the fire gives out light (radiation)

radiator of a car::

- conducts the heat of the engine (conduction)
- water pump circulates hot antifreeze to radiator (convection)
- radiators radiate heat by emitting infrared light (radiation)

antifreeze = a liquid that lowers water's freezing point when added.